



HÁSKÓLI ÍSLANDS

Lokaverkefni til BSc gráðu
í sálfræði

The effects of interletter and line spacing on reading speed, comprehension, and perceived comfortability

Sóley Kristjánsdóttir
Víðir Gunnarsson

Leiðbeinandi: Tómas Kristjánsson, PhD
Júní 2024

SÁLFRÆÐIDEILD

The effects of interletter and line spacing on reading speed, comprehension, and perceived comfortability

Sóley Kristjánsdóttir
Víðir Gunnarsson

Lokaverkefni til BSc-prófs í sálfræði
Leiðbeinandi: Tómas Kristjánsson, PhD

Sálfræðideild
Heilbrigðisvísindasvið Háskóla Íslands
Júní 2024

The effects of interletter and line spacing on reading speed, comprehension, and perceived comfortability.

Ritgerð þessi er 10 eininga lokaverkefni til BSc-prófs
við sálfræðideild á Heilbrigðisvísindasviði Háskóla Íslands

© 2024 Sóley Kristjánsdóttir og Víðir Gunnarsson
Ritgerðina má ekki afrita nema með leyfi höfunda.

Abstract

In the modern world, electronic devices are becoming a bigger part of people's daily lives, and they are increasingly used to seek information, including written material. This has raised curiosity to look at reading off mobile phones. Researchers have looked at reading on different electronic devices with different conditions, for example, wider line length, wider letter spacing, longer letter spacing, and many others. In this current study, the purpose was to look at the influence of letter spacing and line spacing regarding reading speed, comprehension, and experience of reading on a mobile device. The results showed no significant difference between letter spacing or line spacing regarding reading speed or reading comprehension. The analysis showed a significant difference in comfortability for different text conditions. This current study is an important base for continuing studies on what affects reading on mobile phones.

Keywords: reading speed, reading comprehension, letter spacing, line spacing

Preface

We would like to thank Dr. Tómas Kristjánsson for his good guidance, support, and advice. We would also like to thank Davíð Haraldsson for his good advice and help with creating the experiment. We are very grateful for the support, patience, and encouragement of our friends and family during this project. And finally, we would like to thank each other for our pleasant cooperation.

Table of contents

Abstract	4
Preface	5
Introduction	7
Method	10
Participants	10
Material	10
Procedure	11
Data analysis	12
Results	13
Discussion	17
Reference	20

Introduction

Imagine that you are out with your friends and one of them asks you if you read the new article they published the day before. You pick up your phone, and within a few seconds you find the article and read it. Mobiles are convenient to read and find written material quickly and easily. Most people have their phones with them at all times and use it to read a wide range of written material. The mobile is a tool that can be useful for a wide range of activities, and has developed a lot with growing technology.

Evolving technology

The telephone has gone through a lot of development to become the smartphone we know today. The smartphone we know has gone through three major steps in evolution to become what it is now. In the beginning, the smartphone was mainly used for businesses; the smartphone that is most associated with this step is the Blackberry, which was prevalent from 2001-2007 which is when the next step started (Seth, 2023). In January 2007 the iPhone was first introduced by Apple (Kerris & Dowling, 2007) and Google unveiled its Android system in November 2007 (Britannica, 2023). This was the first time that the public had access to smartphones. During this time, applications were added to the smartphones to make them more attractive to the consumer, such as, internet with a browsing as a preference, social websites, camera recordings, and audio recording, to mention a few. The third phase of the smartphone involved bridging the gap between business-centered and public-centered systems. This involved enhancing the display, both in quality and technology, and improving the batteries. The two biggest manufacturers of smartphones are Apple, who developed the iPhone, and Samsung, and they release new phones every year, always with the goal of improving the user experience (Reid, 2018).

Research by the Internet and Mobile Association of India found that 65% of searches made in 2009 were done for educational purposes. There is also a sign of a trend in the popularity of e-readers, which suggests e-learning by reading e-books. E-books are books that you have on your device and can be purchased or downloaded online from anywhere. When compared to paper books, e-books are, in general, cheaper and easier to share, which makes them more attractive than traditional paper books (Kumar, 2011). Using e-books can be environmentally better than paper books, both in terms of trees and CO₂ emissions. A single tree can make around 100 books if they are on average 400 pages long; however, various e-book readers can

have as many as 15.100 books on them, and smartphones with the option of deleting and downloading new books can have an infinite amount of books on them. Making a single e-reader makes on average a CO2 footprint of 168 kg, while a single book produces around 7.5kg. This means you would need to read around 22 books to have the same CO2 footprint (Towler, 2022). However, the biggest reason people read online is convenience. If you wanted to read a book on paper, you would need to figure out which bookstore sells it, then travel there and purchase the book, all before you had a chance to read it. However, when you read a book online, it can be found, purchased, or downloaded and ready to read within minutes.

Computers and other electronic devices are increasingly used in teaching as the technology continues to develop. Students tend to be positive about using electronic devices (Milliner & Cote, 2015). Mobile devices are more preferred by students, rather than desktop PCs when it comes to learning (Lin, 2014). There are all kinds of applications available on mobile devices with teaching material and instructions that can help students to learn, e.g. to read.

Reading

Reading is a complex ability that is important to thrive in a society where a lot of information is presented in written form. The majority of people learn to read in early childhood and will continue to use reading skills throughout their whole life, for enjoyment, to learn new knowledge and many other circumstances. There are many cognitive processes involved in reading (Rayner & Reichle, 2010). Reading can be different depending on the type of text being read. While reading a novel, it is possible to skim through rather quickly and understand the storyline. But when reading scientific articles or other complex texts, more attention is needed to comprehend the text. To research reading comprehension based on a text, it is common to ask questions based on the content of the text (Rayner et al., 2012).

Reading comprehension

Reading comprehension is a process of thinking where the individual selects information or facts and determines the meaning the writer intended. There are several processes in reading comprehension that enable these processes, for example, vocabulary and word recognition (McKee, 2012). The general assumption is that reading comprehension will automatically become worse when moving from paper to any form of electronic device. However, it is risky to make assumptions before researching the subject. A number of researchers have not found

a significant difference in reading comprehension between electronic devices and paper (Margolin et al., 2013; Long & Szabo, 2016; Sackstein et al., 2015; Dundar & Akcayir, 2012). These results indicate that reading comprehension should not change when moving from paper to electronic devices. Meta-analysis by Schwabe et al. (2022) was unable to show a negative association between reading comprehension on electronic devices compared to paper. In fact, reading on electronic devices has shown to have positive effects on reading comprehension (Kaman & Ertem, 2018) however, the effect seems to decrease with long term use.

Reading speed

Reading speed is different between people, some read faster than others. Reading fast has benefits, it can be time saving and can increase how much is read on screen. Faster readers tend to remember more than those that read slow, but that does not mean that individuals that train to read faster will remember more automatically, there are more factors involved (Dyson & Haselgrove, 2001). Factors like letter spacing and line spacing can affect reading speed.

Text layout

Increasing letter spacing can lead to a lower number of letters that can be recognized by a glance in the visual span. Fewer letters in the visual span due to letter spacing leads to slower reading (Chung, 2002; Yu et al., 2007). Increased number of characters per line results in faster and more efficient reading (Kolars et al., 1981; Duchnicky & Kolars, 1983). Optimal letter spacing for dyslexic people is often larger than for other readers (Zorzi et al., 2012), larger letter spacing reduces crowding, which is often a problem with individuals with dyslexia. Larger letter spacing slows down reading for individuals that already read fast, the reading will not go as smoothly with the extra spacing (Korinth et al., 2020).

Individuals have different preferences for text layout. There are countless different ways to influence the material read, that could affect reading speed. Different line length can influence the reading process, but as long as the lines are within reasonable limits, reading can go ahead smoothly (Slattery & Rayner, 2013). There are more factors than line lengths that can influence reading. Line spacing is something that can affect reading. While some might prefer long line spacing, others would rather choose short line spacing. There are a lot of different line spacing that can be used, and there is probably an optimal line spacing as well. Chan et al.

(2014) found that the optimal spacing between lines was 1,5 lines. Although this study might lack generalizability since they used Chinese text.

Reading on different platforms

Reading is different depending on the reading platform. Reading off digital devices is unlike reading from paper. Reading speed varies between platforms, reading can be faster for electronic devices (Hsieh et al., 2016; Sackstein et al., 2015). However, not everyone has found this effect (Siegenthaler et al., 2011).

There can be different explanations as to why larger letter and line spacing will lead to slower reading speed while reading on a mobile phone. Larger letter and line spacing take up more space than smaller spacing, resulting in more screens required to fit in all the text. When there are more screens of texts, reading speed will decrease, due to more time being spent on scrolling. Then on the other hand, smaller letter and line spacing takes up less space and therefore requires fewer screens. Fewer screens lead to faster reading considering less time spent on scrolling.

The current study

Prior studies have researched reading across different electronic devices versus paper, however, there is limited research on reading off mobile phones. In the current study, the purpose is to explore the relationship between different variables, both line spacing and letter spacing and the effect of those variables on reading speed, reading comprehension and subjective experience while reading off mobile phones. The hypothesis are that when the text has both short line letter spacing and line spacing, participants will read the text faster than in the neutral text, and their reading comprehension will not be influenced. Furthermore, participants will find the short letter and short line spacing as comfortable as the other conditions.

Method

Participants

18 participants volunteered in the experiment, 13 of whom were female. The average age was 24,7 ranging from 22-31. The sample was based on convenience. All participants had normal or corrected to normal vision; no participant had dyslexia or learning disabilities. All participants had prior experience reading on mobile devices. Participants received no payment. The appropriate ethics committee gave its approval to the experiment in all respects, and it complied with the Declaration of Helsinki for testing human subjects.

Material

There were two phones used in the experiment: the iPhone 14 Pro with a resolution of 2556 x 1179 and the iPhone 11 with a resolution of 1792 x 828. The application that was used to create and perform the experiment was Figma.

Figma is an online application that can be used to create and make designs for websites, mobile apps, and other digital materials. It is a tool that is used by designers, product managers, and others involved in design processing (What is Figma?, n.d.).

Screen recording was used to document the results. The variables measured were the time taken to read each text, the answers to the questions after each text, and the Likert scale questions about the subjective experience of reading after each text.

Texts

The texts were found on the website *Vísindavefurinn*, which is an Icelandic website that posts various scientific articles or stories, as well as allowing people to ask questions to which they can receive answers. The texts that were used in the experiment were of various subjects, but all had the common aspect of being about abstract objects or people that the average person would not have prior knowledge of to ensure that they would not know answers without reading the texts. There were nine different texts that varied from 157 to 330 words. The texts were, for example, about the great wall of China, the hibernation of bears, and facts about octopuses (see Appendix 1 for an example of a text from the experiment). The questions that followed were about facts found in the texts (see Appendix 2 for an example).

All participants read nine texts. The texts were in the font Inter, and the font size was 14 px. The order of the texts and conditions was counterbalanced. There were three levels of letter

spacing: -1, 0 and 2,5; there were also three levels of spacing between lines 120%, 145% and 170% so there were nine conditions in total that each participant read. The choice of levels of letter spacing and line spacing was decided to maintain practicality and efficiency of reading and to not be too far from normal text display.

Procedure

The experiment took place in a bright room with natural light and minimal distractions. Participants were seated and held the phone in their hands. All participants read a form with information about the experiment and signed informed consent (see Appendix 3).

Participants received instructions for the application and how the procedure would work. The application consisted of text on the screen, then a *back* button in the bottom left corner and a *continue* button in the bottom right corner. Participants read the text and then pressed continue to go to a page that had the questions. Reading comprehension questions were in a multiple-choice format with four possible answers. Participants answered background questions about age, gender, and education. After the instructions, the researcher turned on screen recording and left the room so there would not be any distractions. There was no time limit.

After each text, the participants had to answer five questions that tested their reading comprehension. The questions were made up from the information in the texts; see Appendix 2 for examples of questions translated to English. There were also questions regarding reading subjective experience after each condition to measure the participants' experience of the condition, and they were: How easy was the reading? How comfortable was the text? And, how easily did you maintain attention?

Data analysis

The data file was made in Google Sheets and converted into an Excel file (Microsoft Corporation, 2023) for easy analysis. The data recorded was reading time, percentage of correct answers and the scores from the subjective experience questions. Before the data analysis was performed, the data was examined, and no data was missing. The data analysis was performed in the statistical software SPSS version 28.0 (IMB Corp., 2021).

The first part of the analysis examined the descriptive statistics for the number of participants, the mean and the standard deviation for each variable, the reading speed within

each condition, the reading comprehension depending on the variable, and the comfortability of each text depending on the condition.

A univariate ANOVA was performed for reading speed to examine if there was a difference in reading speed between different text conditions. A univariate ANOVA was also performed for reading comprehension to examine whether there was a difference in reading comprehension depending on the text condition. Also, to look at the difference between comfortability for different text conditions, a univariate ANOVA was performed. Finally, an one-way ANOVA was performed to examine gender differences in reading speed.

Results

Reading speed

In Table 1, there are descriptive statistics for the time taken to read the texts. The shortest time was in the short letter long line spacing with a time of 57,2 seconds, and the longest time was in the long letter long line spacing with a time of 250,2 seconds. The shortest mean was in the text with both short letter spacing, and short line spacing with a time of 116,1 seconds, while the longest mean was in the text with long letter and the long line spacing, with a mean time of 141,7 seconds.

As seen in Table 1, the mean reading speeds of males were faster in all conditions. The results of the one-way ANOVA for reading speed between the genders were significant $p < .001$. These results indicate that there is a difference between the genders regarding reading speed.

Table 1
Descriptive statistics for reading speed

		SS	SM	SL	MS	MM	ML	LS	LM	LL
Male	Mean	93,1	107,5	99,6	113,2	103,9	97,0	106,9	108,9	108,4
	SD	21,9	26,5	18,9	34,3	34,6	28,9	25,2	18,9	24,9
Female	Mean	124,9	136,3	138,4	138,9	147,0	140,3	151,2	140,0	154,4
	SD	31,4	41,6	45,2	46,6	27,7	30,7	51,9	34,6	46,6
Total	Mean	116,1	128,3	127,6	131,8	135,0	128,3	138,9	131,3	141,7

Note. S = short, M = medium, L = long

Results of the univariate ANOVA test showed no significant effect for reading speed regarding letter spacing $F(2, 153) = 0,115$, $p = ,891$, or line spacing $F(2, 153) = 1,483$, $p = ,230$. This suggests that reading speed is not influenced by letter spacing or line spacing. The interaction

effect of reading speed and condition (letter spacing, line spacing) was not significant $F(4, 153) = 0,426, p = ,790$.

Reading comprehension

After each text, the participants answered five questions from each text. The lowest score was in the texts with the variables short letter spacing and long line spacing, and in the long letter spacing short line spacing no correct answers (0%). As seen in Table 2, on at least one occasion all of the variables had the highest score available, which was 100%. The lowest mean score was in the long letter spacing medium line spacing (57,8%) and the highest was in the medium letter spacing short line spacing (78,9%).

Table 2
Descriptive statistics for reading comprehension

	N	Minimum	Maximum	Mean	SD
Answer % SS	18	20%	100%	71,1%	24,9%
Answer % SM	18	40%	100%	77,8%	19,3%
Answer % SL	18	0%	100%	61,2%	30,3%
Answer % MS	18	40%	100%	78,9%	23,2%
Answer % MM	18	20%	100%	66,7%	26,6%
Answer % ML	18	20%	100%	72,2%	19,6%
Answer % LS	18	0%	100%	67,7%	33,6%
Answer % LM	18	20%	100%	57,8%	24,6%
Answer % LL	18	40%	100%	68,9%	23,0%

Note. S = short, M = medium, L = long

Results of the univariate ANOVA test showed no significant effect for reading comprehension regarding letter spacing $F(2, 153) = 0,647, p = ,525$ or line spacing $F(2, 153) = 1,451, p = ,237$. Letter spacing and line spacing do not appear to have an effect on reading comprehension. The

interaction effect was not significant between reading comprehension and condition (letter spacing, line spacing) $F(4, 153) = 1,664, p = ,161$.

Comfortability

Comfortability of reading was measured with Likert scale questions after each text condition. As seen in Table 3, the lowest score in comfortability was found in three different conditions and the highest score was found for all conditions except for two. The highest rated text condition in comfortability was medium letter spacing, long line spacing (4,22).

Table 3
Descriptive statistics for comfortability

	N	Minimum	Maximum	Mean	SD
Comfortability SS	18	2	4	3,28	0,75
Comfortability SM	18	1	5	3,39	1,24
Comfortability SL	18	2	5	3,50	0,86
Comfortability MS	18	2	4	3,50	0,62
Comfortability MM	18	1	5	3,78	1,21
Comfortability ML	18	2	5	4,22	0,80
Comfortability LS	18	1	5	3,39	1,19
Comfortability LM	18	2	5	3,44	0,92
Comfortability LL	18	2	5	3,83	1,04

Note. S = short, M = medium, L = long

The results of the univariate ANOVA were significant for letter spacing $F(2, 153) = 3,113, p = ,047$. These results indicate that comfortability is different depending on letter spacing. Mean for long letter spacing ($M = 3,85$) was significantly different than short letter spacing ($M = 3,39$). Meaning that longer letter spacing made the text more comfortable to read. However, medium letter spacing ($M = 3,54$) did not significantly differ from long or short letter spacing.

The results of the univariate ANOVA for line spacing were marginally significant $F(2, 153) = 2,808$, $p = ,063$. This outcome suggests that different line spacing has an impact on comfortability. Mean for medium line spacing ($M = 3,83$) was marginally significantly different than short line spacing ($M = 3,39$). Meaning that medium line spacing made the text more comfortable to read, than short line spacing. However, long line spacing ($M = 3,56$) did not significantly differ from medium or short line spacing. These results suggest that both letter spacing and line spacing have an effect on comfortability. However, the interaction effect of letter spacing and line spacing is not strong enough to have a significant effect $F(4, 153) = 0,344$, $p = ,855$, therefore, it is not possible to claim whether there is interaction between the letter and line spacing.

Discussion

The purpose of this study was to examine whether there was an optimal way of reading text on a mobile phone that would optimize both the reading speed and the reading comprehension by altering either the letter spacing or line spacing. This was accomplished by altering the variables (letter spacing, line spacing) of the texts. The study was done in cooperation with psychologists that are developing application software, and the texts and questions were in a prototype of that app to make it as relevant to the app design as possible (Souldis, n.d.).

The results of the ANOVA test on letter spacing and line spacing on reading speed were not significant. These results mean that it is not possible to claim that the conditions used affected reading speed of participants. However, these results go against results of prior research (Chung, 2002) showing that shorter distance between letters results in faster reading, and, (Yu et al., 2007), that showed that longer space between letters results in slower reading. However, prior research has shown that increased letter spacing can have a positive effect on people with dyslexia. This could be due to the fact that dyslexic people suffer from crowding when reading, so increasing the interletter spacing can help reduce this crowding effect (Zorzi et al., 2012).

The results of different letter and line spacing were not significant for reading comprehension. These findings mean that reading comprehension was not influenced by the text conditions used in this study. This goes against prior research that found that increased interletter spacing resulted in increased reading comprehension (Dotan & Katzir, 2018).

The results of comfortability were significant for letter spacing, meaning that letter spacing had influence on comfortability. Participants showed a preference to long letter spacing rather than short letter spacing, indicating that they find the long letter spacing more comfortable. This is in concurrence with prior research on dyslexic individuals (Zorzi et al., 2012). This goes against the hypothesis of current study, since it was assumed that there would not be a difference in comfortability between text conditions. Medium letter spacing did not have a significant difference in comfortability compared to the other conditions.

Medium line spacing was marginally significant from short line spacing for comfortability, suggesting that medium line spacing has some effect on reading comfortability, even though the result did not reach the statistical significant threshold. These results suggest that participants seem to prefer medium line spacing over the other variables, but additional

research would be needed to confirm these effects of line spacing. Long line spacing did not have a significant nor marginally difference of the other conditions.

Interesting results that were not part of the hypothesis, were that male participants read significantly faster than female participants. These results mean that males read faster than females. However, considering the small sample size, it would be intriguing for future studies to examine gender differences in a larger sample to see if it would lead to the same results.

The experiment had a few issues that were related to the use of the application Figma during the experiment. The application is still in development, so it still has some glitches and bugs. The sample size was also quite small, with only 18 participants; that potentially had an influence on the results. The study also had an uneven gender distribution, with 13 female and 5 male participants. A uniform distribution between genders could yield more valid results; this is further demonstrated by the results, which found significant differences in reading speed between genders.

Even though the results for letter spacing and line spacing were not significant for reading speed or reading comprehension, this study is still useful for further research on reading on mobile phones or other electronic devices. Future research could look more into gender differences in reading speed, perhaps with different texts and other text conditions. Furthermore, it would be possible to look at text variables that have more distinguishable differences, since in the current study the conditions were relatively similar to standard text. This study implies how important it is to research reading of mobile phones and expand knowledge on how letter spacing and line spacing can affect the reading experience when reading on a mobile phone.

References

- Bragason, S. H. (2003, May 27). *Hvað getur þú sagt mér um Júpíter?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=3453>
- Britannica, T. Editors of Encyclopaedia. (2023, December 27). *Android*. Britannica. <https://www.britannica.com/technology/Android-operating-system>
- Chan, A. H. S., Tsang, S. N. H., & Ng, A. W. Y. (2014). Effects of line length, line spacing, and line number on proofreading performance and scrolling of Chinese text. *Human factors*, 56(3), 521-534. <https://doi.org/10.1177/0018720813499368>
- Chung, S. T. L. (2002). The effect of letter spacing on reading speed in central and peripheral vision. *Visual Psychophysics and Physiological Optics*, 43(4), 1270-1276. <https://iovs.arvojournals.org/article.aspx?articleid=2200181#89592014>
- Dotan, S., & Katzir, T. (2018). Mind the gap: Increased inter-letter spacing as a means of improving reading performance. *Journal of Experimental Child Psychology*, 174, 13–28. <https://doi.org/10.1016/j.jecp.2018.04.010>
- Duchnicky, R. L., & Kolars, P. A. (1983). Readability of Text Scrolled on Visual Display Terminals as a Function of Window Size. *Human Factors*, 25(6), 683–692. <https://doi.org/10.1177/001872088302500605>
- Dundar, H., & Akcayir, M. (2012). Tablet vs. Paper: The Effect on Learners' Reading Performance. *International Electronic Journal of Elementary Education*, 4(3), 441–450. <https://iejee.com/index.php/IEJEE/article/view/188>
- Dyson, M. C., & Haselgrove, M. (2001). The influence of reading speed and line length on the effectiveness of reading from screen. *International Journal of Human-Computer Studies*, 54(4), 585-612. <https://doi.org/10.1006/ijhc.2001.0458>
- Halldórsson, J. M. (2001, October 16). *Hvaða dýr sefur mest?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=1911>
- Halldórsson, J. M. (2003, April 16). *Hvað getið þið sagt mér um sæhesta?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=3346>
- Halldórsson, J. M. (2005, December 23). *Hvað getið þið sagt mér um fjallaljón?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=5509>
- Hsieh, Y. C., Kuo, C. T., & Lin, H. (2016). The effect of screen size of mobile devices on reading efficiency. *Design for aging*. Zhou, J. & Salvendy, G. (eds.), (pp. 435-445). https://doi.org/10.1007/978-3-319-39943-0_42
- IMB Corp. (2021). *IMB SPSS Statistics for Macintosh* (Version 28.0).

- Ingólfsson, Á. H. (2014, December 4). *Hvers konar tónlistarmaður var Mozart og hvernig kynnti hann sér verk barrokkmeistaranna?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=68487>
- Jóhannesson, G. T. (2006, January 19). *Hvernig maður var Sveinn Björnsson forseti og hvað afrekaði hann?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=5575>
- Kaman, S., & Ertem, I. S. (2018). The effect of digital texts on primary students' comprehension, fluency, and attitude. *Eurasian Journal of Educational Research*, 18(76), 147-164. <https://doi.org/10.14689/ejer.2018.76.8>
- Kerris, N., & Dowling, S. (2007, January 9). *Apple reinvents the phone with iPhone*. Apple. <https://www.apple.com/newsroom/2007/01/09Apple-Reinvents-the-Phone-with-iPhone/>
- Kolers, P. A., Duchnick, R. L., & Ferguson, D. C. (1981). Eye movement measurement of readability of CRT displays. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 23(5), 517–527. <https://doi.org/10.1177/001872088102300502>
- Korinth, S. P., Gerstenberger, K., & Fiebach, C. J. (2020). Wider Letter-Spacing Facilitates Word Processing but Impairs Reading Rates of Fast Readers. *Frontiers in Psychology*, 11, 444–444. <https://doi.org/10.3389/fpsyg.2020.00444>
- Kumar, M. (2011). Impact of the evolution of smart phones in education technology and its application in technical and professional studies: Indian perspective. *International Journal of Managing Information Technology*, 3(3), 39-49. <https://doi.org/10.5121/ijmit.2011.3304>
- Lin, C. (2014). Learning English reading in a mobile-assisted extensive reading program. *Computers and Education*, 78, 48–59. <https://doi.org/10.1016/j.compedu.2014.05.004>
- Long, D., & Szabo, S. (2016). E-readers and the effects on students' reading motivation, attitude and comprehension during guided reading. *Cogent Education*, 3(1), 1197818-. <https://doi.org/10.1080/2331186X.2016.1197818>
- Margolin, S. J., Driscoll, C., Toland, M. J., & Kegler, J. L. (2013). E-readers, computer screens, or paper: Does reading comprehension change across media platforms? *Applied Cognitive Psychology*, 27(4), 512–519. <https://doi.org/10.1002/acp.2930>
- McKee, S. (2012). Reading comprehension, What we know: A review of research 1995 to 2011. *Language Testing in Asia*, 2(1), 45-58. <https://doi.org/10.1186/2229-0443-2-1-45>
- Microsoft Corporation. (2023). *Microsoft Excel* (version 16.80). <https://office.microsoft.com/excel>
- Milliner, B., & Cote, T. (2015, December). *One year of extensive reading on mobile devices: engagement and impressions*. EUROCALL Conference, Padova, Italy. <https://doi.org/10.14705/rpnet.2015.000366>
- Pétursdóttir, U., Blöndal, B., & Sigurðsson, J. (2002, 11. April) *Hvað getið þið sagt mér um Finnland?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=2286>

- Rayner, K., & Reichle, E. D. (2010). Models of the reading process. *Wiley Interdisciplinary Reviews. Cognitive Science*, 1(6), 787–799. <https://doi.org/10.1002/wcs.68>
- Rayner, K., Pollatsek, A., Ashby, J., & Clifton, C. (2012). *The psychology of reading*. Psychology Press. <https://doi.org/10.4324/9780203155158>
- Reid, A. J. (2018). *The smartphone paradox*. Springer International Publishing. https://doi.org/10.1007/978-3-319-94319-0_2
- Sackstein, S., Spark, L., & Jenkins, A. (2015). Are e-books effective tools for learning? Reading speed and comprehension : iPad® vs. paper. *South African Journal of Education*, 35(4), 1-14. <https://doi.org/10.15700/saje.v35n4a1202>
- Schwabe, A., Lind, F., Kosch, L., & Boomgaarden, H. G. (2022). No negative effects of reading on screen on comprehension of narrative texts compared to print: A meta-analysis. *Media Psychology*, 25(6), 779-796. <https://doi.org/10.1080/15213269.2022.2070216>
- Seth, S. (2023, March 17). *BlackBerry: A story of constant success & failure*. Investopedia. <https://www.investopedia.com/articles/investing/062315/blackberry-story-constant-success-failure.asp#:~:text=The%20golden%20period%20of%202001,expanded%20into%20the%20consumer%20market.>
- Siegenthaler, E., Wurtz, P., Bergamin, P., & Groner, R. (2011). Comparing reading processes on e-ink displays and print. *Displays*, 32(5), 268-273. <https://doi.org/10.1016/j.displa.2011.05.005>
- Sigurðardóttir, M. B. (2005, November 10). *Hvað hafa kolkrabbar marga arma?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=5396>
- Slattery, T. J., & Rayner, K. (2013). Effects of intraword and interword spacing on eye movements during reading: Exploring the optimal use of space in a line of text. *Attention, Perception & Psychophysics*, 75(6), 1275-1292. <https://doi.org/10.3758/s13414-013-0463-8>
- Souldis. (n.d.) <https://souldis.com/>
- Sverrisdóttir, H., & Andersson, U. (2002, May 29). *Hvað er Kínamúrinn gamall og langur? Sést hann frá tunglinu?* Vísindavefurinn. <https://www.visindavefur.is/svar.php?id=2438>
- Towler, B. (2022, May 31). *E-readers vs books: Which are better for the environment?* Commercial Waste. <https://commercialwaste.trade/e-readers-vs-books-better-environment/>
- What is Figma? (n.d.). *Figma Learn*. <https://help.figma.com/hc/en-us/articles/14563969806359-What-is-Figma->

- Yu, D., Cheung, S.-H., Legge, G. E., & Chung, S. T. L. (2007). Effect of letter spacing on visual span and reading speed. *Journal of Vision*, 7(2), 1-10. <https://doi.org/10.1167/7.2.2>
- Zorzi, M., Barbiero, C., Facoetti, A., Lonciari, I., Carrozzi, M., Montico, M., Bravar, L., George, F., Pech-Georgel, C., & Ziegler, J. C. (2012). Extra-large letter spacing improves reading in dyslexia. *Proceedings of the National Academy of Sciences - PNAS*, 109(28), 11455–11459. <https://doi.org/10.1073/pnas.1205566109>

Appendix 1

Kolkrabbar kallast á ensku octopus og á latínu Octopoda, en bein íslensk þýðing á þessum orðum myndi vera áttfætlingur eða átta arma dýr. Þetta er mjög lýsandi fyrir útlit kolkrabba þar sem þeir hafa átta arma, en reyndar geta armarnir stundum verið færri þar sem eitt af varnarviðbrögðum kolkrabba er að aflima sig. Kolkrabbar teljast til fylkingar lindýra (Mollusca) líkt og til dæmis sniglar (Gastropoda), nökkvar (Polyplacophora), samlokur (Bivalvia) og sætennur (Scaphopoda). Þeir tilheyra flokki svokallaðra höfuðfætlinga (Cephalopoda) en innan þess flokks eru, auk kolkrabba, perlusnekkjur (e. nautilus), blekfiskar (e. cuttlefish) og smokkfiskar (e. squid). Til eru um 289 tegundir kolkrabba innan 14 ætta og tveggja undirættbálka. Þetta er því margbreytilegur hópur sem lifir á fjölbreytilegum sjávarbúsvæðum, þótt flestar tegundir sé að finna á grunnsævi, á eða við kóralrif. Armar kolkrabbans eru venjulega þaktir sogskálum. Þessar sogskálar eru mjög næmar og á þeim eru efnanemar sem gera kolkröbbum kleift að "smakka" það sem þeir snerta. Armarnir eru því helsta skynfæri kolkrabbans jafnframt því að gera þeim kleift að synda eða skríða á sjávarbotninum. Annað helsta skynfæri kolkrabba er sjónin, en þeir hafa mjög góða sjón þó að þeir sjái ekki í lit. Búkur kolkrabba líkist einna mest stóru höfði sem gengur út frá miðpunkti armanna. Ólíkt öðrum lindýrum hafa flestir þeirra hvorki ytri né innri skel, en eini harði hluti þeirra er goggur sem minnir um margt á páfagauksogg. Þetta gerir kolkrabba einkar sveigjanlega og auðveldar þeim að smjúga á milli steina eða inn í þröngar rifur til að flýja rándýr. Önnur varnarviðbrögð sem kolkrabbar hafa eru felulitir, blekpokar og sjálfsaflimun (e. autotomy). Kolkrabbar hafa sérstakar húðfrumur sem gera þeim bæði kleift að skipta litum og endurspegla ljósi. Þennan eiginleika nota þeir til þess að renna saman við umhverfi sitt og fela sig þannig, hafa samskipti sín á milli eða sem aðvörun. Sem dæmi má nefna að tegundir af ættkvíslinni Hapalochlaena eru mjög eitraðar og þegar þær verða fyrir áreiti verða þær skærgular að lit með bláa hringi eða línur.

Appendix 2

Examples of questions from the experiment translated to English:

How many kilograms of fat do bears accumulate in a week?

What is the size of the hippocampus in centimeters?

How many earth years does it take Jupiter to go around the sun?

Appendix 3

Upplýsingar um rannsókn

Heiti rannsóknar

Rannsókn á lesskilning og leshraða af símaskjá.

Markmið og tilgangur rannsóknar

Markmið rannsóknarinnar er að mæla lesskilning og leshraða. Upplýsingum verður safnað á meðal þátttakenda í tilrauna herbergi.

Öflun þátttakenda og þáttökuskilyrði

Þáttökuskilyrði er að vera með rétta sjón eða leiðréttu og að vera á aldrinum 20-35 ára. Gert er ráð fyrir að safna svörum frá 18 þátttakendum.

Hvað felst í þátttöku í rannsókninni?

Í þátttöku þinni felst að lesa 9 texta og svara spurningum eftir hvern þeirra. Gert er ráð fyrir að tilraun taki um 1 klukkustund. Tekið skal fram að kjósir þú að taka ekki þátt í rannsókninni mun það ekki hafa neinar afleiðingar. Við gerum okkur grein fyrir að þátttaka þín krefst tíma og fyrirhafnar af þinnar hálfu og yrðum við því mjög þakklát fyrir þátttöku þína.

Kostnaður/greiðslur

Þátttaka mun hvorki fela í sér kostnað fyrir þig né færðu greitt fyrir þátttöku þína.

Mögulegar áhættur og óþægindi við þátttöku

Áhætta við þátttöku beinist fyrst og fremst að trúnað við þátttakendur og verndun persónugreinanlegra auðkenna. Við merkjum spurningalista ekki með nafni þannig að ekki verður hægt að rekja svör til einstakra þátttakenda. Engar upplýsingar munu koma fram í kynningu á niðurstöðum rannsóknar, sem hægt verður að rekja til einstakra þátttakenda. Áhætta þín af þátttöku í rannsókninni er því hverfandi.

Mögulegur ávinningur

Þú hefur ekki beinan hag af þátttöku í rannsókninni. Við vonum þó að þú sjáir hag í því að vita að þátttaka þín geti mögulega átt þátt í því að auka skilning í ..

Hætt við þátttöku

Það er þitt val hvort þú tekur þátt í rannsókninni eða ekki. Þér er frjálst að hafna þátttöku eða hætta í rannsókninni á hvaða stigi sem er, án útskýringa og mun það ekki hafa áhrif á þátttöku þína í hópnum.

Trúnaður við þátttakendur

Nafn þitt kemur hvergi fram í rannsókninni svo svörin verður ekki hægt að rekja til þín. Allar upplýsingar sem þú veitir í rannsókninni, verða meðhöndlaðar samkvæmt ströngustu reglum um trúnað og nafnleynd. Farið verður að íslenskum lögum varðandi persónuvernd, vinnslu og eyðingu frumgagna. Rannsóknargögn verða varðveitt á öruggum stað hjá rannsóknaraðila á meðan á rannsókn stendur og öllum gögnum verður eytt að rannsókn lokinni. Gögnin verða eingöngu notuð af rannsækendum. Niðurstöðurnar verða notaðar til að auka innsýn og skilning á lestri af farsíma. Niðurstöðurnar verða allar ópersónugreinanlegar.

Ábyrgðarmaður og aðrir rannsakendur

Tómas Kristjánsson, ábyrgðarmaður rannsóknarinnar.

Sálfræðingur á Kvíðameðferðarstöðinni, Suðurlandsbraut 4, 108 Reykjavík.

Aðjúntkt við Sálfræðideild Háskóla Íslands

netfang: tomask@kms.is.

Með fyrirfram þökk,

Tómas Kristjánsson, ábyrgðarmaður rannsóknar

Ég hef lesið og kynnt mér upplýsingarnar um þessa rannsókn og rétt minn til að hætta þátttöku. Ég veiti hér með samþykki mitt fyrir þátttöku í rannsókninni og notkun gagnanna sem úr henni kemur.